

LAB EXPERIMENTS – ITA04 STATISTICS WITH R PROGRAMMING

1. Create numeric, character, and logical vectors and display type and content.

Program:

```
# Numeric vector
```

```
num <- c(10, 20, 30)
```

```
# Character vector
```

```
char <- c("A", "B", "C")
```

```
# Logical vector
```

```
logi <- c(TRUE, FALSE, TRUE)
```

```
print(num)
```

```
print(typeof(num))
```

```
print(char)
```

```
print(typeof(char))
```

```
print(logi)
```

```
print(typeof(logi))
```

output:

```
> # Numeric vector
> num <- c(10, 20, 30)
>
> # Character vector
> char <- c("A", "B", "C")
>
> # Logical vector
> logi <- c(TRUE, FALSE, TRUE)
>
> print(num)
[1] 10 20 30
> print(typeof(num))
[1] "double"
>
> print(char)
[1] "A" "B" "C"
> print(typeof(char))
[1] "character"
>
> print(logi)
[1] TRUE FALSE TRUE
> print(typeof(logi))
[1] "logical"
```

2. Create labeled matrices (5×4 , 3×3 , 2×2) filled by row/column

Program

```
# 5×4 matrix (filled by row)
```

```
m1 <- matrix(1:20, nrow=5, ncol=4, byrow=TRUE)
```

```
rownames(m1) <- c("R1","R2","R3","R4","R5")
```

```
colnames(m1) <- c("C1","C2","C3","C4")
```

```
print(m1)
```

```
# 3×3 matrix (filled by column)
```

```
m2 <- matrix(1:9, nrow=3, byrow=FALSE)
```

```
rownames(m2) <- c("A","B","C")
```

```
colnames(m2) <- c("X","Y","Z")
```

```
print(m2)
```

```
# 2×2 matrix
```

```
m3 <- matrix(1:4, nrow=2, byrow=TRUE)
```

```
rownames(m3) <- c("R1","R2")
```

```
colnames(m3) <- c("C1","C2")
```

```
print(m3)
```

output:

```

> # 5x4 matrix (filled by row)
> m1 <- matrix(1:20, nrow=5, ncol=4, byrow=TRUE)
> rownames(m1) <- c("R1", "R2", "R3", "R4", "R5")
> colnames(m1) <- c("C1", "C2", "C3", "C4")
>
> print(m1)
  C1 C2 C3 C4
R1  1  2  3  4
R2  5  6  7  8
R3  9 10 11 12
R4 13 14 15 16
R5 17 18 19 20
>
> # 3x3 matrix (filled by column)
> m2 <- matrix(1:9, nrow=3, byrow=FALSE)
> rownames(m2) <- c("A", "B", "C")
> colnames(m2) <- c("X", "Y", "Z")
>
> print(m2)
  X Y Z
A 1 4 7
B 2 5 8
C 3 6 9
>
> # 2x2 matrix
> m3 <- matrix(1:4, nrow=2, byrow=TRUE)
> rownames(m3) <- c("R1", "R2")
> colnames(m3) <- c("C1", "C2")
>
> print(m3)
  C1 C2
R1  1  2
R2  3  4

```

3. Create and display a 3D array

Program

```
arr <- array(1:24, dim=c(2,3,4))
```

```
print(arr)
```

output:

```

> arr <- array(1:24, dim=c(2,3,4))
> print(arr)
, , 1
      [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6

, , 2
      [,1] [,2] [,3]
[1,]    7    9   11
[2,]    8   10   12

, , 3
      [,1] [,2] [,3]
[1,]   13   15   17
[2,]   14   16   18

, , 4
      [,1] [,2] [,3]
[1,]   19   21   23
[2,]   20   22   24

```

4. Create arrays with dimension names and print a specific element

Program

```
v <- 1:12
```

```

arr <- array(v,
             dim=c(2,3,2),
             dimnames=list(
               Row=c("R1","R2"),
               Col=c("C1","C2","C3"),
               Table=c("T1","T2")
             ))

```

```
print(arr)
```

```
print(arr["R2","C3","T2"])
```

output:

```
>
> v <- 1:12
>
> arr <- array(v,
+             dim=c(2,3,2),
+             dimnames=list(
+                 ROW=c("R1","R2"),
+                 Col=c("C1","C2","C3"),
+                 Table=c("T1","T2")
+             ))
>
> print(arr)
, , Table = T1

    Col
Row  C1 C2 C3
R1   1  3  5
R2   2  4  6

, , Table = T2

    Col
Row  C1 C2 C3
R1   7  9 11
R2   8 10 12

>
> print(arr["R2","C3","T2"])
[1] 12
```

5. Create and manipulate factor variables

Program

```
heights <- factor(c("Short","Medium","Tall","Medium","Tall"))
```

```
print(heights)
```

```
print(levels(heights))
```

```
letters_factor <- factor(sample(LETTERS[1:5],10,replace=TRUE))
```

```
print(letters_factor)
```

```
summary(letters_factor)
```

output:

```

>
> print(arr["R2","C3","T2"])
[1] 12
>
>
> heights <- factor(c("Short","Medium","Tall","Medium","Tall"))
>
> print(heights)
[1] Short Medium Tall Medium Tall
Levels: Medium Short Tall
> print(levels(heights))
[1] "Medium" "Short" "Tall"
>
> letters_factor <- factor(sample(LETTERS[1:5],10,replace=TRUE))
>
> print(letters_factor)
[1] C C C C B A C C D A
Levels: A B C D
> summary(letters_factor)
A B C D
2 1 6 1

```

6. Create an R list containing vectors, matrices and functions

Program

```
vec <- c(1,2,3)
```

```
mat <- matrix(1:4,2,2)
```

```
mylist <- list(
  Vector=vec,
  Matrix=mat,
  Function=sum
)
```

```
print(mylist)
```

```
mylist$Function(vec)
```

output:

```

>
> vec <- c(1,2,3)
>
> mat <- matrix(1:4,2,2)
>
> mylist <- list(
+   Vector=vec,
+   Matrix=mat,
+   Function=sum
+ )
>
> print(mylist)
$Vector
[1] 1 2 3

$Matrix
      [,1] [,2]
[1,]    1    3
[2,]    2    4

$Function
[1] 650

```

7. Write R programs for basic tasks: Factors of a number, generate a vector of 10 random integers

between -50 and 50, print numbers 1–100 with FizzBuzz logic.

(a). Find factors of a number

Program

```
num <- 24
```

```
for(i in 1:num)
```

```
{
```

```
  if(num %% i == 0)
```

```
    print(i)
```

```
}
```

Output:

```

> num <- 24
>
> for(i in 1:num)
+ {
+   if(num %% i == 0)
+     print(i)
+ }
[1] 1
[1] 2
[1] 3
[1] 4
[1] 6
[1] 8
[1] 12
[1] 24

```

(b). Generate 10 random integers between -50 and 50

Program

```
set.seed(100)
```

```
x <- sample(-50:50,10)
```

```
print(x)
```

output:

```

>
> set.seed(100)
>
> x <- sample(-50:50,10)
>
> print(x)
[1] 23 38 27 -28 35 19 -47 4 45 -44

```

(c). FizzBuzz (1 to 100)

Program

```
for(i in 1:100)
```

```
{
```

```
  if(i %% 3 == 0 && i %% 5 == 0)
```

```
    print("FizzBuzz")
```

```
  else if(i %% 3 == 0)
```



```
    print("Fizz")
else if(i %% 5 == 0)
    print("Buzz")
else
    print(i)
}
```

Output:

```
> for(i in 1:100)
+ {
+   if(i %% 3 == 0 && i %% 5 == 0)
+     print("FizzBuzz")
+   else if(i %% 3 == 0)
+     print("Fizz")
+   else if(i %% 5 == 0)
+     print("Buzz")
+   else
+     print(i)
+ }
[1] 1
[1] 2
[1] "Fizz"
[1] 4
[1] "Buzz"
[1] "Fizz"
[1] 7
[1] 8
[1] "Fizz"
[1] "Buzz"
[1] 11
[1] "Fizz"
[1] 13
[1] 14
[1] "FizzBuzz"
```